Title, authors, year	Dataset	Application area	Method / model	Headline metrics
Research on classification method of multimodal magnetic resonance images of Alzheimer's disease based on generalized convolutional neural networks Qin et al., 2023, J Biomed Eng	ADNI; 91 subjects (44 AD, 47 NC); sMRI + rs-fMRI	AD vs NC classification	gCNN: 3D HA-ResUNet (sMRI) + U-GCN (fMRI brain network); feature fusion, binary PSO feature selection, ML classifier	ACC 88.89%, SEN 100%, SPE 77.78%, G-mean 88.19% (best single modality, U-GCN: ACC 83.33%)
s2MRI-ADNet: an interpretable deep learning framework integrating Euclidean-graph representations of Alzheimer's disease solely from structural MRI Song et al., 2024, MAGMA	ADNI, AIBL, EDSD, OASIS; 3,377 subjects; sMRI only	NC vs AD classification, within and across databases	Dual channel: GMV maps with CNN-Transformer + regional radiomics similarity network (R2SN) with GraphSAGE and MLP	Intra-database ACC 92.1%, AUC 0.962; inter-database ACC 91.4%, AUC 0.938
A Graph Convolutional Network Based on Univariate Neurodegeneration Biomarker for Alzheimer's Disease Diagnosis Qu et al., 2023, IEEE JTEHM	ADNI, Aβ+ subjects; sMRI cortical thickness (FreeSurfer, left hemisphere, 34 ROIs)	AD vs CU and AD vs MCI classification	GCN on a population graph (age, gender, APOE edges) with univariate neurodegeneration biomarker (UNB) node features and an attention module	Validation ACC 93.90% (AD vs CU), 82.05% (AD vs MCI)
A Hybrid Deep Learning Framework to Predict Alzheimer's Disease Progression Using Generative Adversarial Networks and Deep Convolutional Neural Networks SinhaRoy & Sen, 2024, Arab J Sci Eng	ADNI-derived 4-class set of 2D sMRI slices; 6,400 images	4-stage AD classification	DCGAN (synthetic images) + SRGAN (super-resolution) + compact 2D CNN	ACC 99.7% (AD vs NL AUC 1.000)
An Alzheimer's Disease classification network based on MRI utilizing diffusion maps for multi-scale feature fusion in graph convolution Yang et al., 2024, Math Biosci Eng	ADNI; 338 AD, 422 MCI, 532 CN; sMRI ROI features (116 AAL regions)	AD / CN / MCI pairwise and 3-class classification	Population graph; diffusion maps at multiple neighbour counts; metric-learned adjacency; self-attention fusion	ACC: AD-CN 0.887, AD-MCI 0.705, CN-MCI 0.825, 3-class 0.671
Graph-Based Multimodal Fusion Framework with Correlation-Aware Learning for Alzheimer's Disease Prediction Li et al., 2026, Big Data Min Anal	ADNI; 417 subjects (103 CN, 103 EMCI, 105 LMCI, 106 AD); sMRI + SNP genetics	CN vs AD, CN vs MCI, MCI vs AD classification	Brain-region graph + gene-ontology graph; GCN with pooling; cross-attention + correlation-aware learning (CAL) module	CN vs AD: ACC about 0.82, AUC 0.82; CN vs MCI: ACC 0.773; MCI vs AD: ACC 0.747
Longitudinal Graph Neural Networks for Prognostic Modeling of Alzheimer's Disease Progression Ashimgaliyev et al., 2026, AAIML	ADNI; about 600 MCI subjects; longitudinal sMRI + FDG-PET / amyloid-PET features	Predicting MCI-to-AD conversion within 5 years	Patch-based 3D CNN embeddings as nodes in a spatiotemporal graph (82 ROIs); GAT (spatial) + GRU with temporal attention	AUC 0.89, C-index 0.84, BAC 86%, SEN 85%, SPE 87%
A multilayered framework for diagnosis and classification of Alzheimer's disease using transfer learned Alexnet and LSTM Goyal, Rani & Singh, 2024, Neural Comput Appl	ADNI T1 MRI; 200 subjects (60 AD, 73 MCI, 67 CN); 5,980 2D slices	3-class (AD / MCI / CN) and pairwise classification	Pre-trained AlexNet features, then LSTM, then transfer-learned dense layers; GAN augmentation of training data	With GAN: ACC 96.83% (3-class), 98.13% (AD-CN), 99.38% (AD-MCI), 99.37% (CN-MCI)

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Machine Learning and Deep Learning for Alzheimer's Disease Diagnosis: A Survey of MRI, PET, Multimodal Fusion, Transformers, Graph Neural Networks, and Explainable AI Ahmad, Tarannum & Uddin, 2026, Am J Adv Technol Eng Solut	Survey, no dataset of its own; covers ADNI, OASIS-3, AIBL	Review of ML/DL for AD diagnosis (MRI, PET, multimodal)	Narrative review organised by traditional ML, CNN, multimodal fusion, transformers, GNN and explainable AI	None (no experiments). Concludes multimodal usually beats single-modality; transformers and GNNs promising but not proven superior
Graph neural networks in Alzheimer's disease diagnosis: a review of unimodal and multimodal advances Ali, Piana, Pardini & Garbarino, 2025, Front Neurosci	73 reviewed studies; mostly ADNI, also TADPOLE, OASIS, UK Biobank, ABIDE	Review of GNN-based AD diagnosis (NC/AD, NC/MCI, MCI/AD, progression)	Review organised by GNN pipeline: graph construction, convolution, pooling, prediction, explainability	None of its own. Best reported 99.3% NC/AD (TADPOLE, multimodal); multimodal studies had higher median accuracy than unimodal
Development of framework by combining CNN with KNN to detect Alzheimer's disease using MRI images Lanjewar, Parab & Shaikh, 2023, Multimed Tools Appl	Kaggle 4-class Alzheimer's MRI set; 6,400 images (ND 3,200, VMD 2,240, MID 896, MOD 64)	4-stage AD classification	4-block CNN as feature extractor feeding a KNN (k = 3); compared with ResNet50, VGG16, MobileNetV2	Mean ACC 99.58% (K = 20), F1 99.43%, MCC 99.31%